

Project Title

Steel Plant Production & Quality Performance Analysis

Project Type: End-to-End Data Analytics Project

This project focuses on analyzing steel plant production and quality data to evaluate production performance, identify high-defect areas, and provide data-driven insights for operational improvement.

Business Problem

A steel manufacturing plant generates large volumes of production and operational data across different departments, machines, shifts, products, and production dates.

However, raw production data alone does not clearly show where production performance is strong or where quality-related problems are occurring.

The business needs to answer the following questions:

- How much total production has been achieved?
- Which departments contribute the most to production?
- Which machines have higher production and defect quantities?
- Which departments have a higher defect rate?
- Which shifts experience higher defect rates?
- Which products have more quality issues?
- How does production and defect rate change over time?
- Which areas require further investigation and improvement?

The purpose of this project is to analyze the available production data and convert it into meaningful business insights using SQL, Python, and Power BI.

Project Objectives

The main objectives of this project are:

1. Analyze the overall production performance of the steel plant.
2. Compare production across different departments.
3. Analyze machine-wise and shift-wise production performance.
4. Calculate the total defect quantity and overall defect rate.

5. Identify departments with higher defect rates.
 6. Identify machines contributing to higher defect quantities or defect rates.
 7. Analyze product-wise and shift-wise quality performance.
 8. Analyze monthly production and defect trends.
 9. Compare production and quality performance across different years.
 10. Develop interactive Power BI dashboards for production and quality analysis.
 11. Identify key business problem areas that require further investigation.
 12. Provide data-driven recommendations based on the analysis.
-

4. Dataset Description

The dataset contains steel plant production and operational records collected across different departments, machines, shifts, operators, and production dates.

Each record represents a production-related observation containing information about production quantity, energy consumption, process parameters, quality status, defects, and downtime.

Main Columns

Column	Description
Production_ID	Unique identifier for each production record
Production_Date	Date on which production was recorded
Department_Name	Name of the production department
Shift_Name	Production shift such as A, B, or C
Machine_ID	Unique machine identifier
Operator_ID	Identifier of the operator
Product_Name	Name of the product produced
Production_Quantity_Tons	Quantity of production in tons
Energy_Consumption_MWh	Energy consumed during production
Process_Temperature_C	Process temperature in Celsius
Process_Pressure_Bar	Process pressure in bar
Quality_Status	Quality result/status of the production
Defect_Quantity_Tons	Quantity of defective material in tons
Downtime_Minutes	Production or machine downtime
Downtime_Reason	Reason for the recorded downtime

Dataset Analysis

The dataset is analyzed at different levels:

- **Department level**
- **Shift level**
- **Monthly level**
- **Yearly level**

The analysis focuses mainly on two business areas:

Production Performance

How much was produced and where was production highest or lowest?

Quality Performance

Where are defects occurring and which areas have a higher defect rate?

The defect rate is calculated using:

$$\text{Defect Rate (\%)} = (\text{Total Defect Quantity} / \text{Total Production Quantity}) \times 100$$

Tools Used

The project uses multiple data analytics tools to perform the complete analysis workflow.

- ✓ **SQL Server**
- ✓ **Python**
 - **Pandas**
 - **Matplotlib**
 - **Seaborn**
- ✓ **Power BI**

SQL Analysis

Total Production

```
SELECT SUM(Production_Quantity_Tons) AS Total_Production_Tons  
  
FROM steel_plant_combineddata;
```

Results		Messages
	Total_Production_Tons	
1	57024244.68	

Department-wise Production

```
SELECT  
    Department_Name,  
    SUM(Production_Quantity_Tons) AS Total_Production_Tons  
FROM steel_plant_combineddata  
GROUP BY Department_Name  
  
ORDER BY Total_Production_Tons DESC;
```

Results		Messages
	Department_Name	Total_Production_Tons
1	Coke Oven	9584528.19
2	Power Plant	9551723.74
3	Steel Melting Shop	9549033.02
4	Maintenance	9536233.70
5	Blast Furnace	9463194.10
6	Rolling Mill	9339531.93

PRODUCT-WISE PRODUCTION

```
SELECT  
    Product_Name,  
    Cast(SUM(Production_Quantity_Tons)as decimal(18,2)) AS Total_Production_Tons  
FROM steel_plant_combineddata  
GROUP BY Product_Name  
  
ORDER BY Total_Production_Tons DESC;
```

Results Messages		
	Product_Name	Total_Production_Tons
1	Billet	9579165.90
2	Hot Rolled Coil	9554726.69
3	Wire Rod	9528416.43
4	TMT Bar	9515654.87
5	Cold Rolled Coil	9427041.38
6	Steel Plate	9419239.41

Year Wise Energy Consumption

```

SELECT
    YEAR(Production_Date) AS Production_Year,
    Cast(SUM(Energy_Consumption_MWh)as decimal(18,2)) AS Total_Energy_MWh
FROM steel_plant_combineddata
GROUP BY YEAR(Production_Date)

ORDER BY Production_Year;

```

Results Messages		
	Production_Year	Total_Energy_MWh
1	2024	14420795.19
2	2025	19778502.71

Defect Analysis

```

SELECT
    Department_Name,

    CAST(SUM(Production_Quantity_Tons) AS DECIMAL(18,2))
    AS Total_Production_Tons,
    CAST(SUM(Defect_Quantity_Tons) AS DECIMAL(18,2))
    AS Total_Defects_Tons,
    CAST(
        SUM(Defect_Quantity_Tons) * 100.0 /
        NULLIF(SUM(Production_Quantity_Tons), 0)
        AS DECIMAL(18,2)
    ) AS Defect_Rate_Percent

FROM steel_plant_combineddata
GROUP BY Department_Name
ORDER BY Defect_Rate_Percent DESC;

```

Results		Messages		
	Department_Name	Total_Production_Tons	Total_Defects_Tons	Defect_Rate_Percent
1	Maintenance	9536233.70	381949.37	4.01
2	Blast Furnace	9463194.10	378499.40	4.00
3	Coke Oven	9584528.19	383447.00	4.00
4	Rolling Mill	9339531.93	372269.67	3.99
5	Power Plant	9551723.74	380976.49	3.99
6	Steel Melting Shop	9549033.02	380524.60	3.98

SHIFT-WISE QUALITY ANALYSIS

```

SELECT
    Shift_Name,
    Quality_Status,
    COUNT(*) AS Total_Records,
    cast(SUM(Production_Quantity_Tons)as decimal(18,2)) AS Total_Production_Tons,
    cast(SUM(Defect_Quantity_Tons)as decimal(18,2) )AS Total_Defect_Tons
FROM steel_plant_combineddata
GROUP BY
    Shift_Name,
    Quality_Status
ORDER BY
    Shift_Name,

    Total_Production_Tons DESC;

```

	Shift_Name	Quality_Status	Total_Records	Total_Production_Tons	Total_Defect_Tons
1	A	Passed	45588	12539816.28	497884.18
2	A	Rejected	11748	3242730.65	130193.21
3	A	Rework	11485	3146300.43	126350.92
4	B	Passed	46545	12823299.65	514098.97
5	B	Rework	11809	3228540.46	130067.61
6	B	Rejected	11375	3142843.71	126008.31
7	C	Passed	46114	12681826.49	506287.97
8	C	Rejected	11493	3174714.98	126196.35
9	C	Rework	11123	3044172.03	120579.01

DOWNTIME REASON ANALYSIS

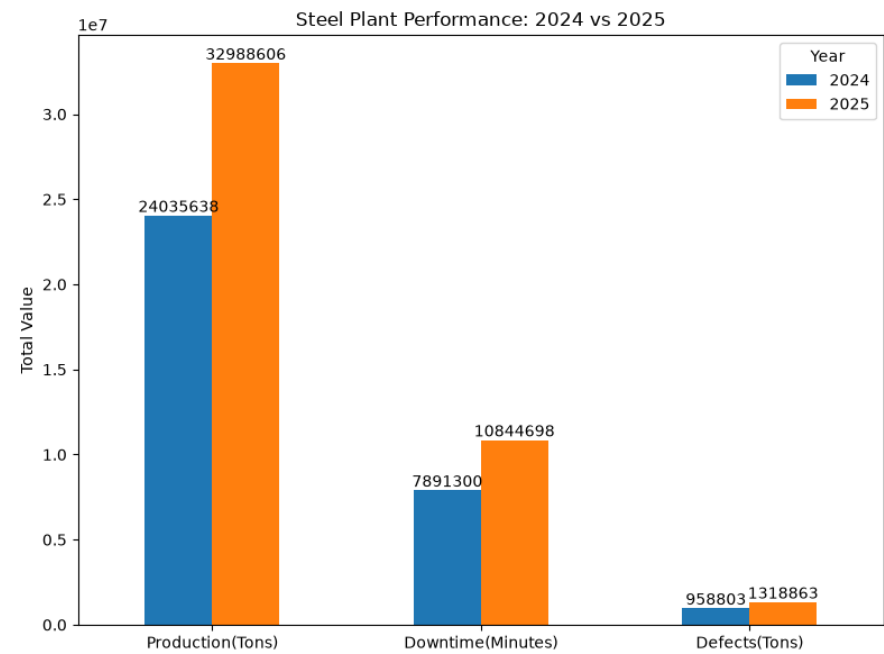
```
SELECT
    Downtime_Reason,
    SUM(Downtime_Minutes) AS Total_Downtime_Minutes
FROM steel_plant_combineddata
GROUP BY Downtime_Reason

ORDER BY Total_Downtime_Minutes DESC;
```

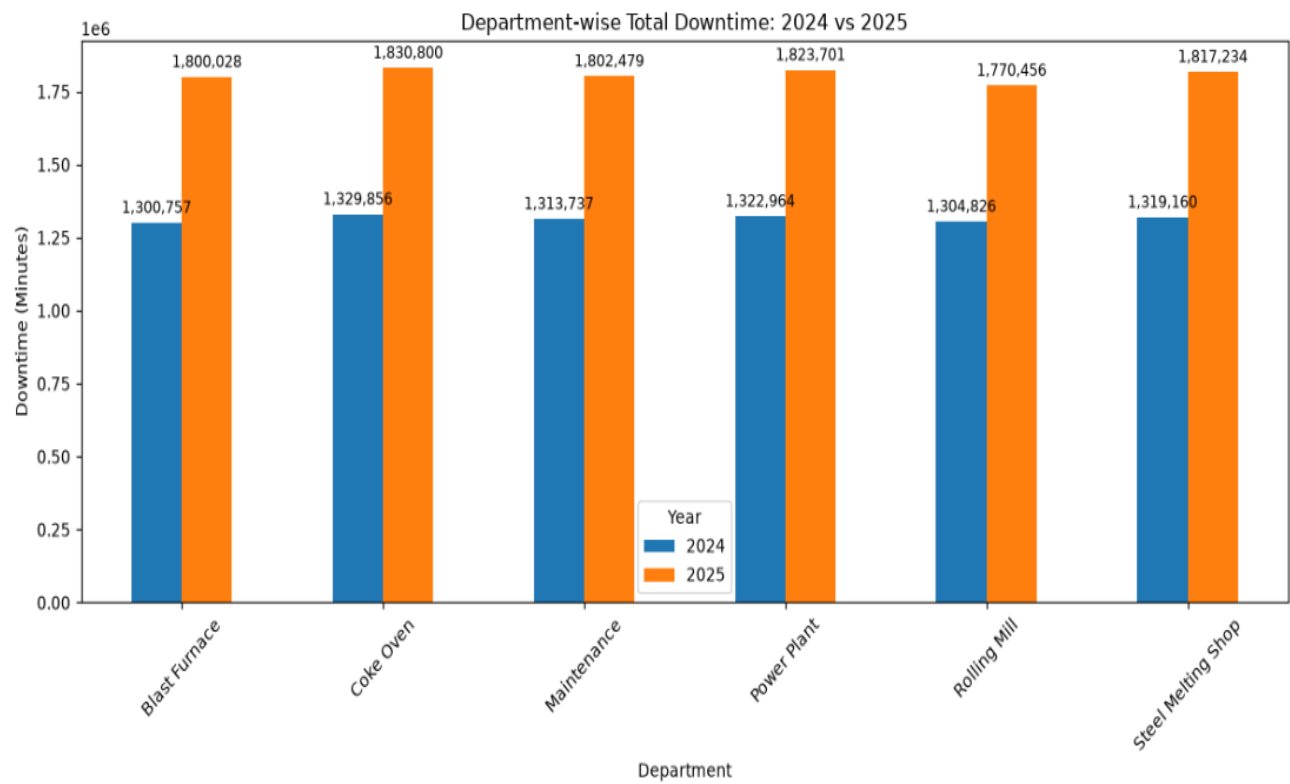
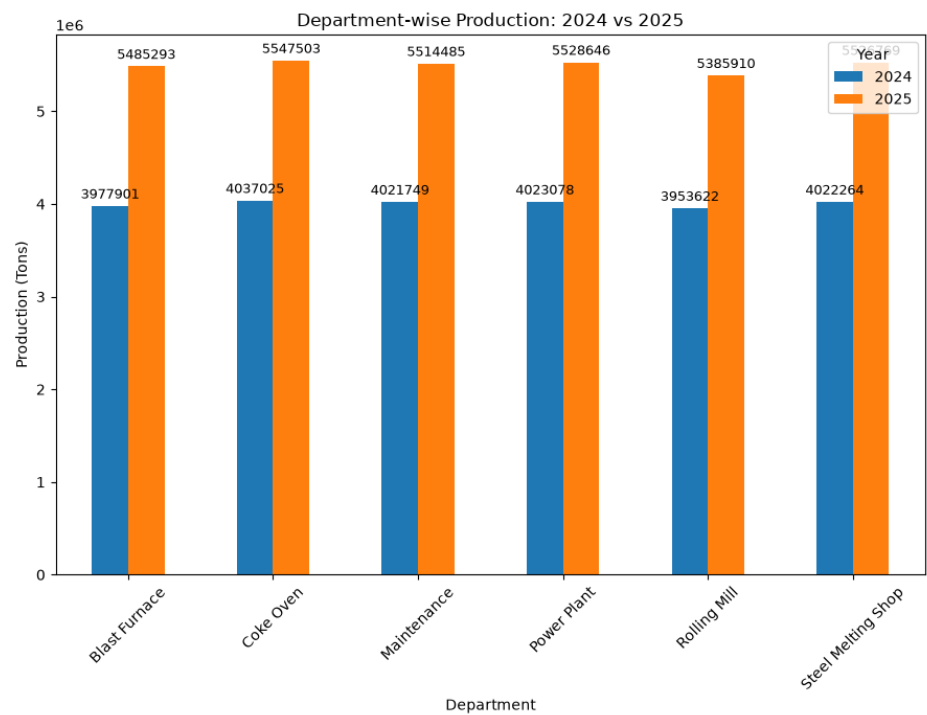
	Downtime_Reason	Total_Downtime_Minutes
1	Process Issue	3774394
2	Machine Failure	3751507
3	Power Failure	3750911
4	Material Shortage	3742659
5	Maintenance	3716527
6	No Downtime	0

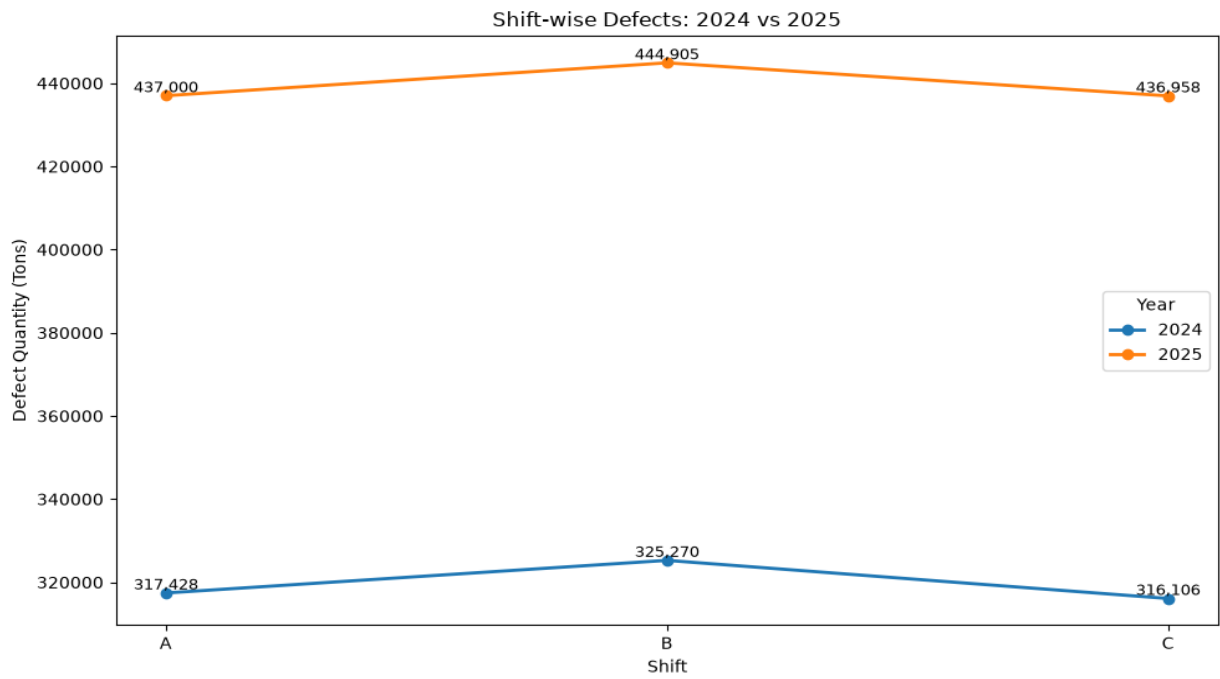
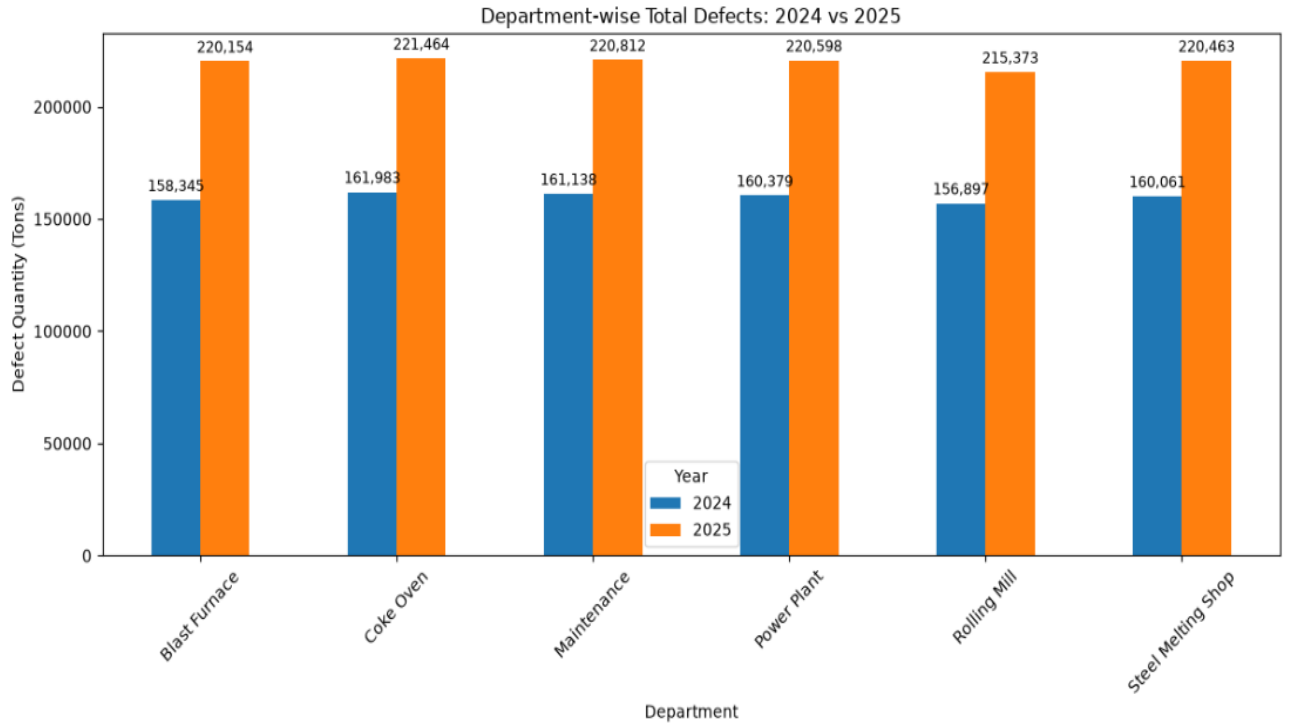
EDA

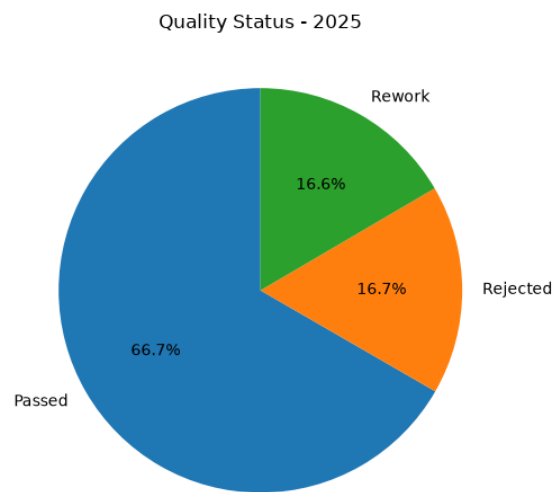
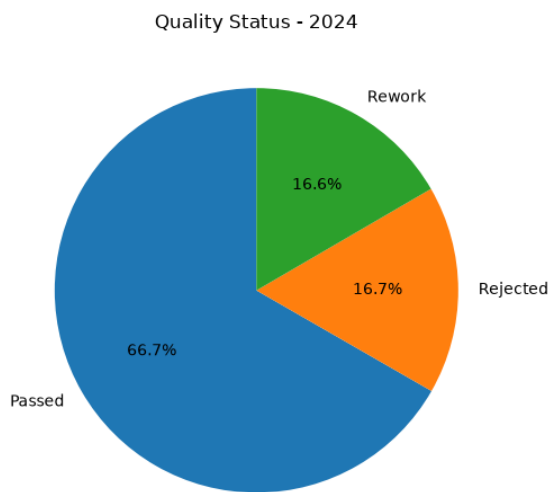
Overall Production Analysis



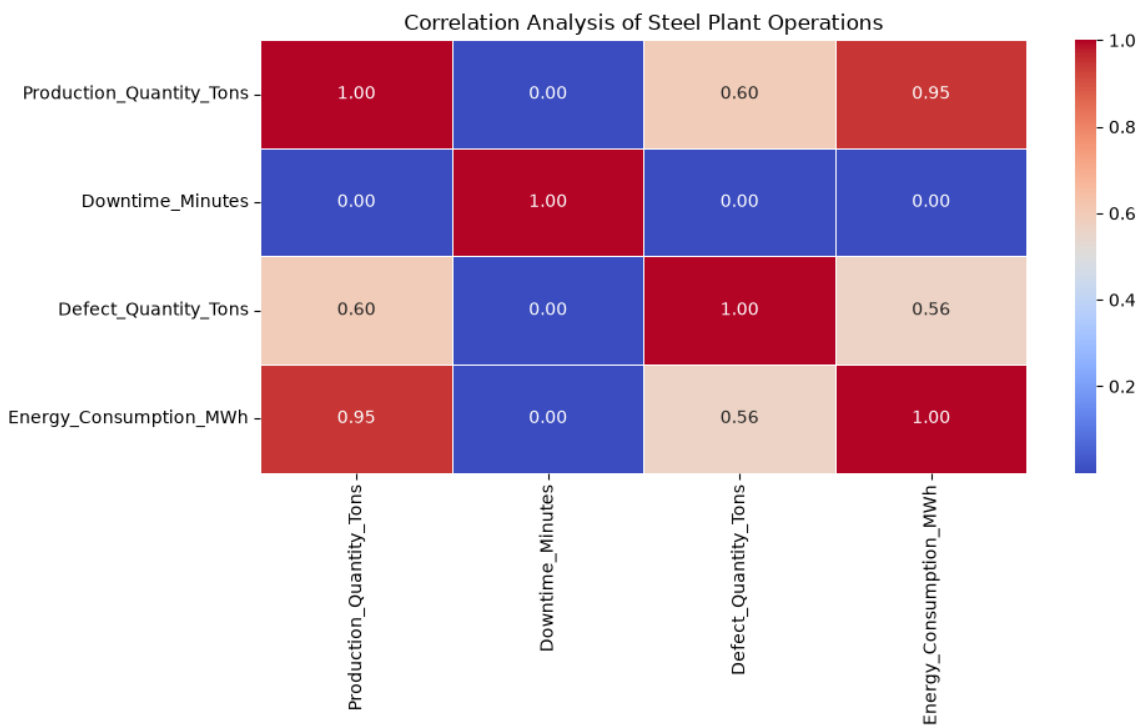
Department-wise Analysis



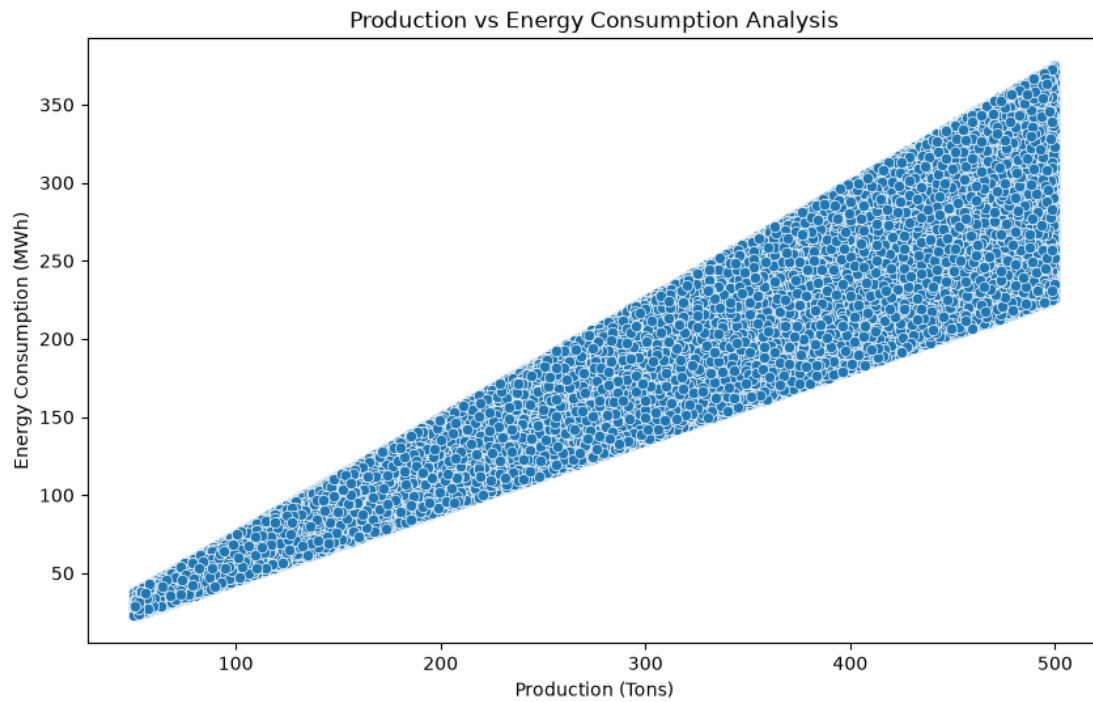




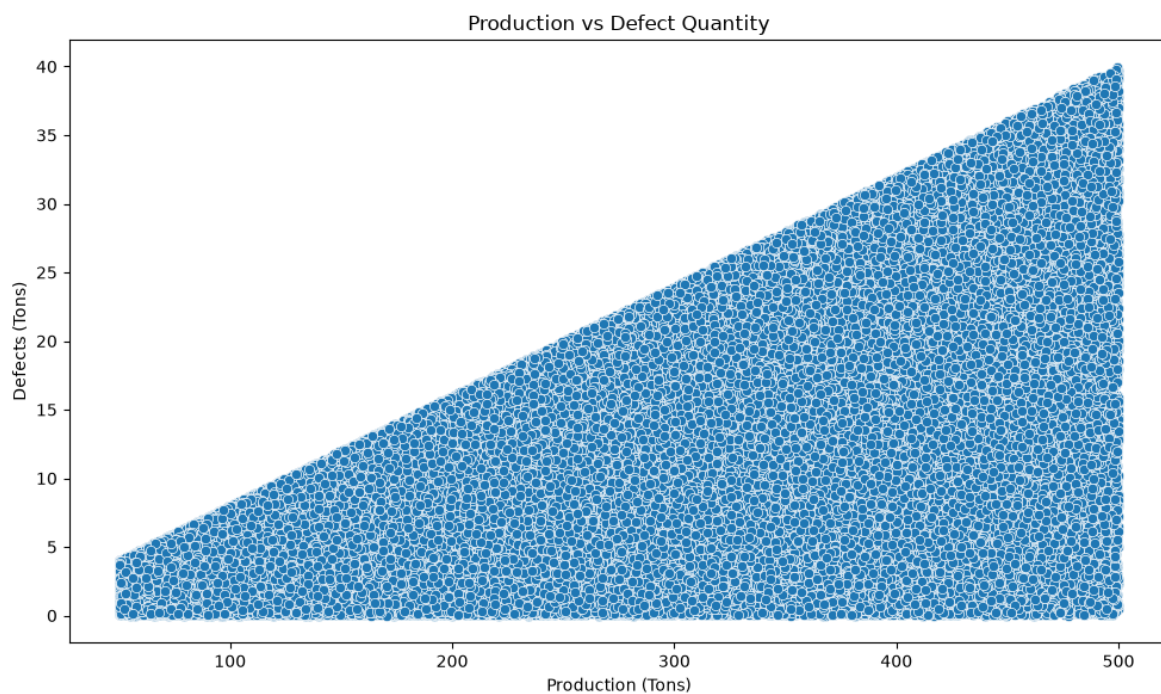
Correlation Analysis



Production ↑ → ***Energy Consumption*** ↑

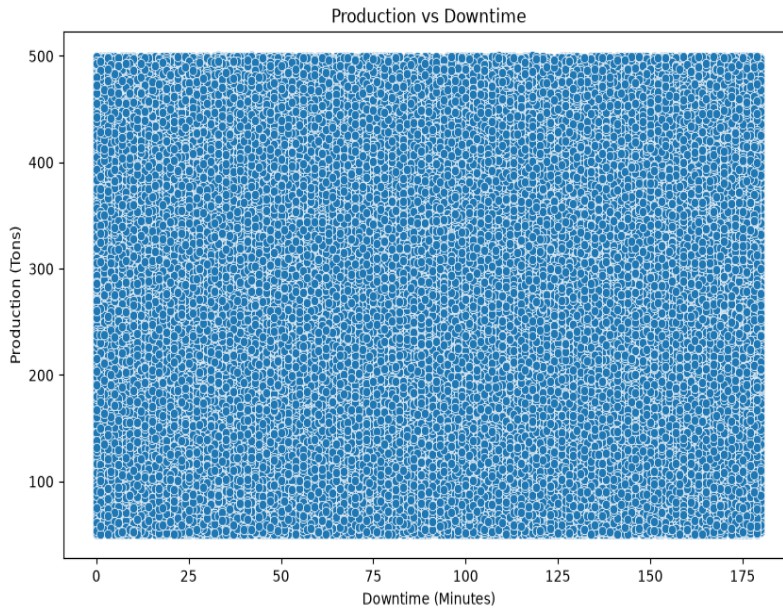


Production ↑ → ***Defects*** ↑



Defects ↑ → Energy Consumption ↑

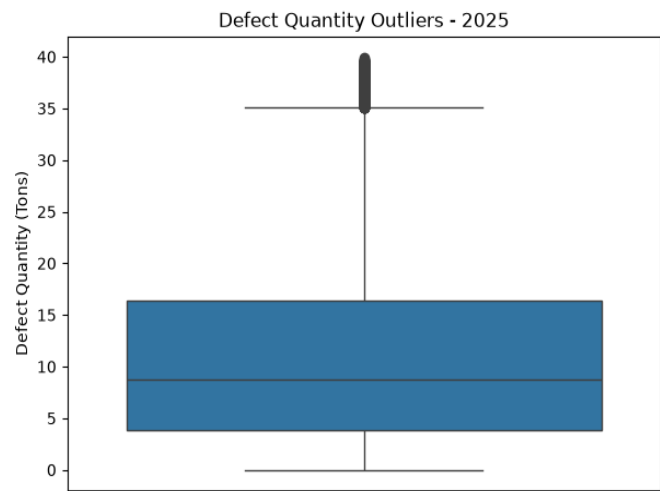
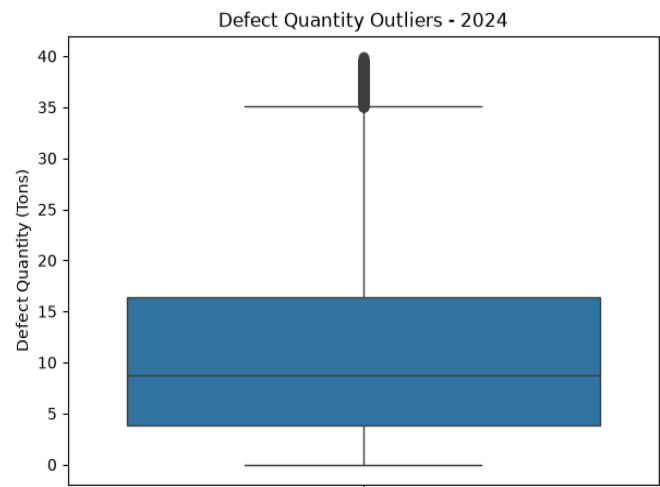
Downtime ↑ → No clear relationship with Production



Downtime ↑ → No clear relationship with Defects

Downtime ↑ → No clear relationship with Energy Consumption

Outlier Analysis



Combine 2024 and 2025 outliers

Q1: 3.88 tons
Q3: 16.38 tons
IQR: 12.5 tons
Lower Limit: -14.870000000000001 tons
Upper Limit: 35.129999999999995 tons
Outlier starts above: 35.129999999999995 tons
Total Outliers: 1843

Outlier Analysis of Defect Quantity

Machine Wise Outliers

Machine_ID	Outlier_Points Outlier_Tons	
MCH-019	45	1658.86
MCH-056	45	1652.85
MCH-053	43	1591.42
MCH-022	43	1560.99
MCH-054	42	1547.52
MCH-058	40	1477.22
MCH-042	38	1390.36
MCH-034	38	1384.00
MCH-005	38	1382.53
MCH-033	37	1356.60
MCH-015	37	1346.71
MCH-025	37	1347.97
MCH-004	36	1313.54
MCH-045	35	1288.30
MCH-023	35	1289.54
MCH-021	35	1274.06
MCH-011	35	1288.61
MCH-049	35	1280.81
MCH-037	34	1254.17
MCH-047	33	1204.95

Operator Wise outliers

Operator_ID	Outlier_Points Outlier_Tons	
OP-109	12	442.60
OP-196	12	445.52
OP-108	11	406.79
OP-170	10	360.30
OP-071	10	360.66
OP-249	10	369.12
OP-462	10	364.43
OP-144	10	363.35
OP-532	10	371.70
OP-009	9	327.33
OP-539	9	330.28
OP-330	9	327.08
OP-492	9	332.89
OP-291	9	334.65
OP-271	9	334.01
OP-039	9	328.14
OP-279	9	320.47
OP-518	8	288.50
OP-066	8	298.76
OP-496	8	298.92

From the defect analysis of 2024 and 2025 combined data, some machines were found to generate more than 35tons of defect quantity repeatedly.

The shift-wise and operator-wise analysis was also performed to understand whether the higher defect values were concentrated in any particular shift or operator. The analysis helps to identify the machines where repeated high defect quantity is occurring.

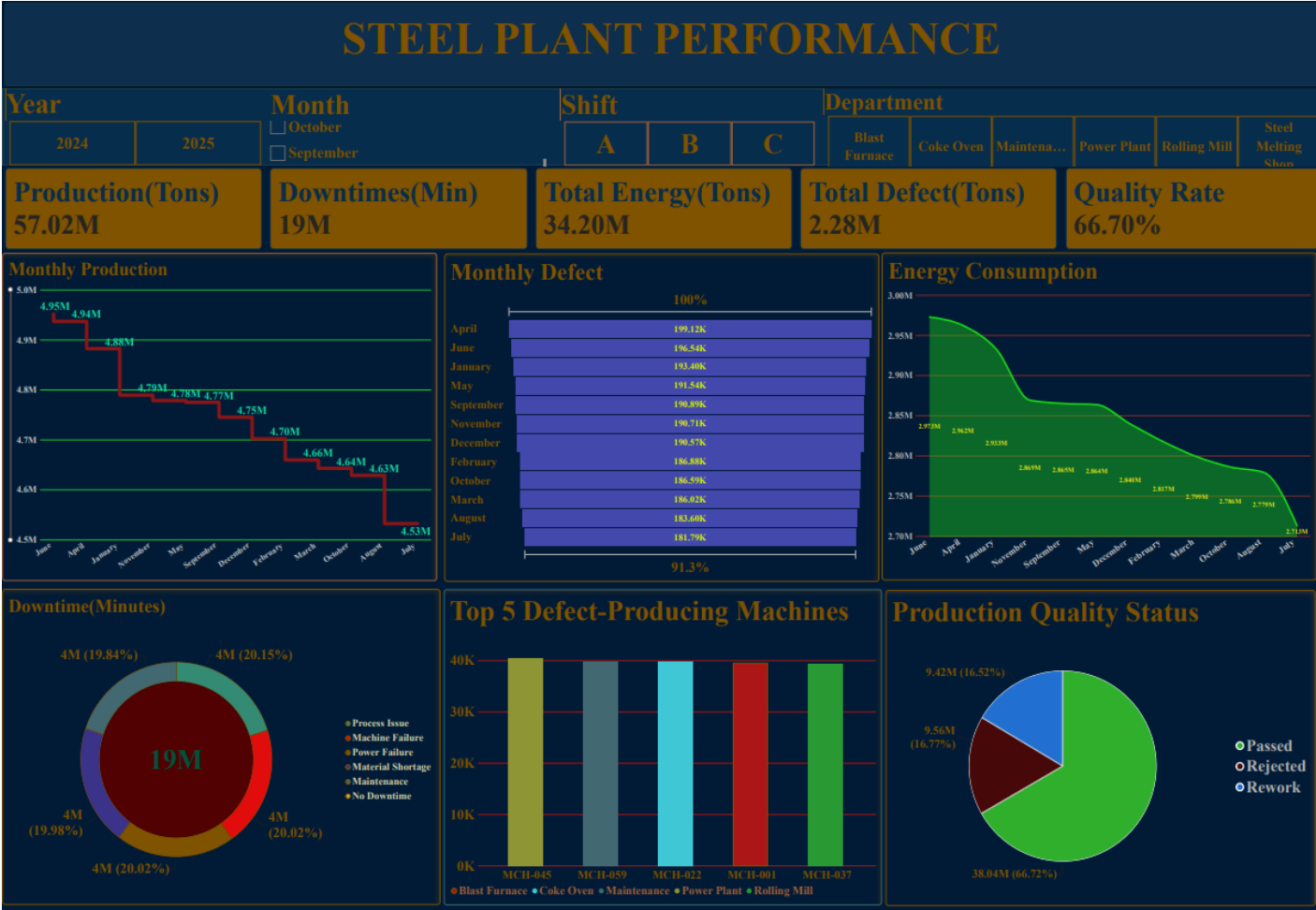
These machines should be checked and investigated by the production and quality teams to understand the possible reasons for higher defect generation. The investigation can include machine condition, maintenance history, production process, raw material quality, and operating conditions.

The identified machines should not be considered as faulty only based on the outlier analysis. The outlier results are used as a warning area for further investigation and monitoring.

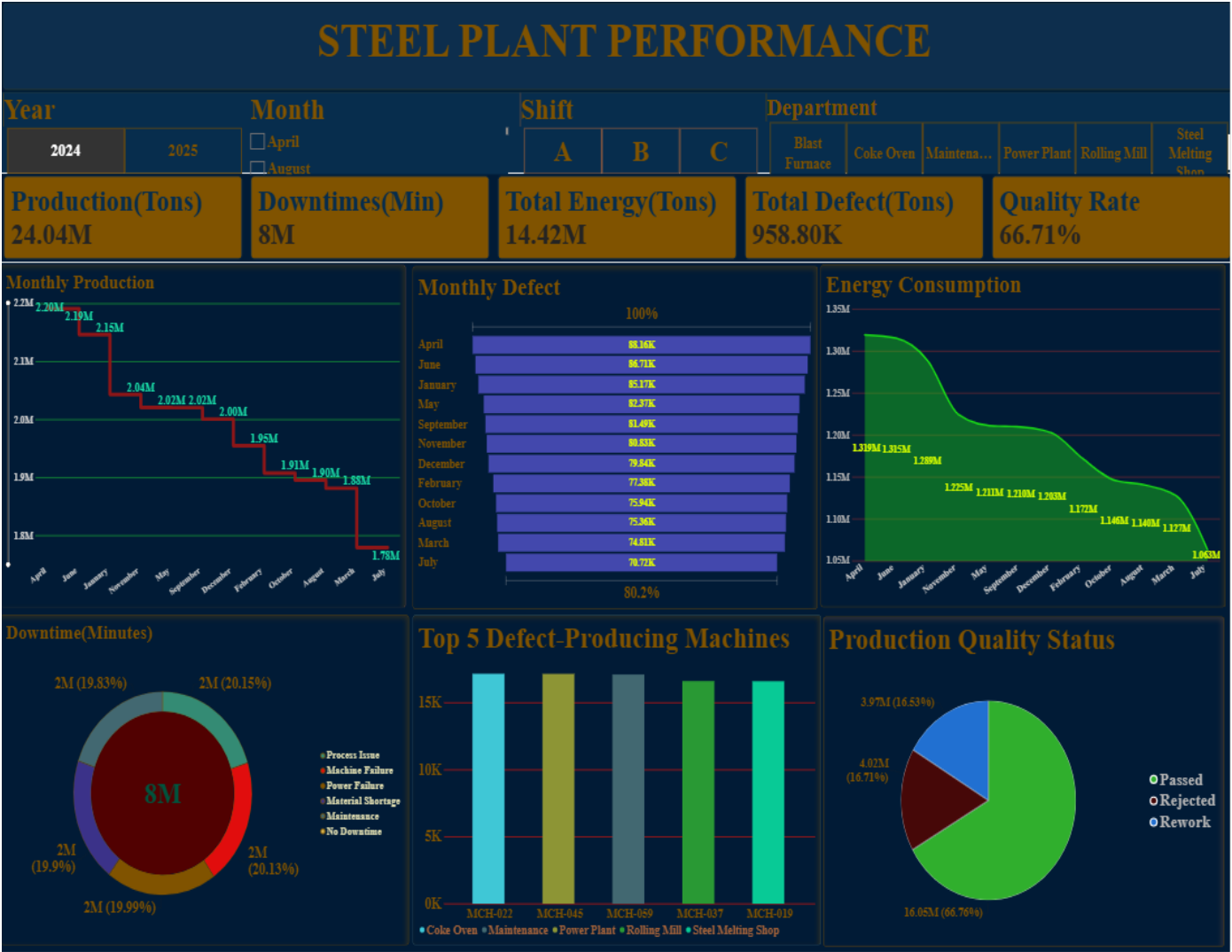
Power BI Dashboard

An interactive Power BI dashboard was developed to provide a consolidated view of steel plant production and quality performance.

Overall Production & Quality Dashboard



2024



STEEL PLANT PERFORMANCE

Year

2024

2025

Month

☐ April

☐ August

Shift

A

B

C

Department

Blast Furnace

Coke Oven

Maintena...

Power Plant

Rolling Mill

Steel Melting Shop

Production(Tons)

32.99M

Downtimes(Min)

11M

Total Energy(Tons)

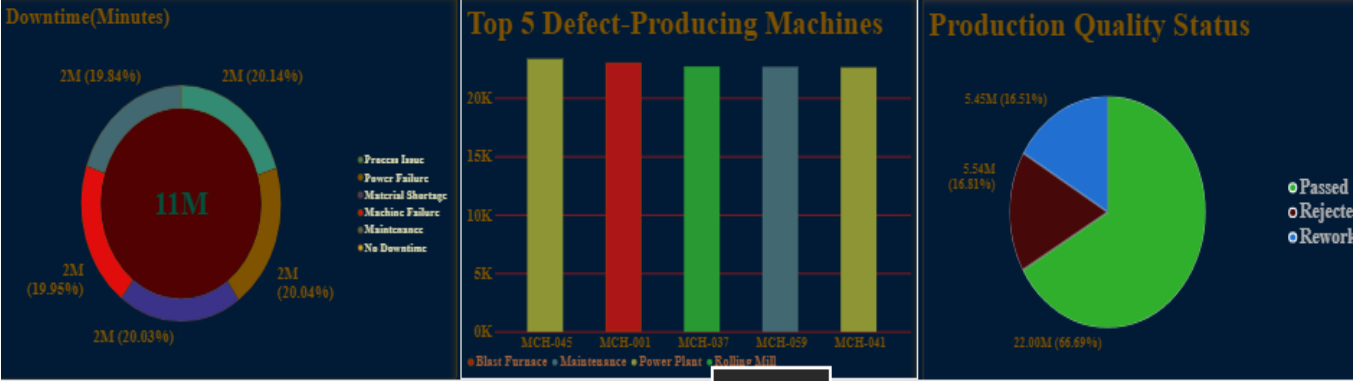
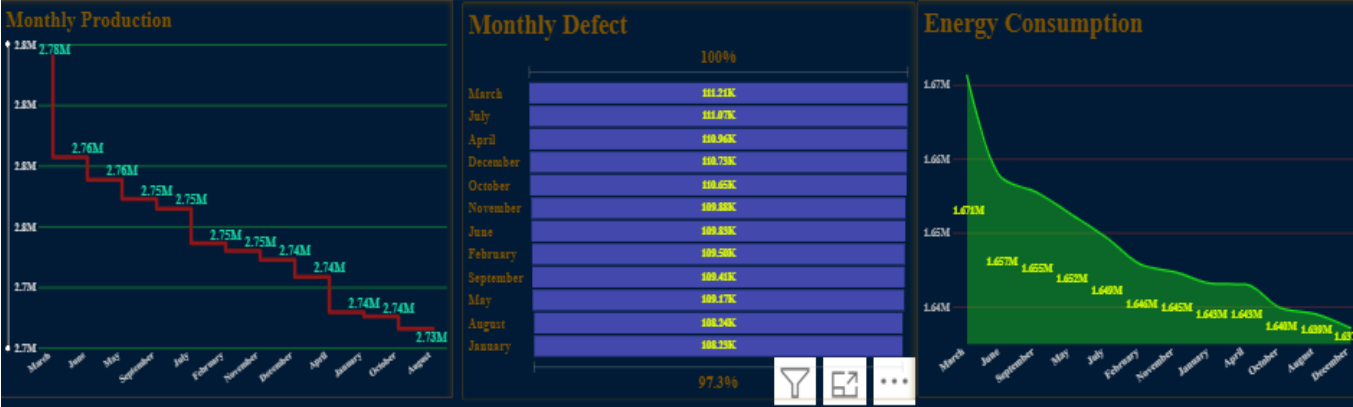
19.78M

Total Defect(Tons)

1.32M

Quality Rate

66.68%



Year

2024

2025

Month

☐ April

☐ August

Shift

A

B

C

Department

Blast Furnace

Coke Oven

Maintena...

Power Plant

Rolling Mill

Steel Melting Shop

13. Conclusion

This project provided a comprehensive analysis of steel plant production and quality performance using SQL, Python, and Power BI.

The analysis helped evaluate total production and compare performance across departments, machines, shifts, and products. Defect quantity and defect rate were analyzed to identify areas where quality issues were relatively higher. Monthly and yearly analysis was also performed to understand production and defect trends over time.

The analysis highlighted specific departments, machines, shifts, and products that require further investigation based on their production and quality performance. This helps in focusing attention on areas where operational or quality-related issues may be occurring.

The Power BI dashboard brings all major insights into a single interactive view, making it easier to monitor production, defects, defect rates, and performance across different dimensions.

Overall, this project demonstrates how raw manufacturing data can be transformed into useful business insights. The findings can support data-driven monitoring, help identify potential quality problem areas, and provide a foundation for further investigation and operational improvement.